

Proximal Policy Optimization for User Association in Hybrid LiFi/WiFi Indoor Networks

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Abstract—Hybrid Light-Fidelity (LiFi) and Wireless-Fidelity (WiFi) indoor networking has been envisioned as a promising technology to alleviate radio frequency (RF) spectrum crunch to accommodate the ever-increasing data rate demand in indoor scenarios. The hybrid LiFi/WiFi indoor networks can utilize the advantages of fast data transfer from LiFi and wider coverage of WiFi, thus complementing well with each of these technologies and increasing the network performance compared to the standalone networks. This work investigates the user-access point (AP) association problem to achieve better network-wide spectral efficiency. We first derive a mathematical formulation of the sum data rate maximization problem by determining the AP selection for each user in indoor networks with either static or mobile users. It is shown that the formulated network control problem is a mixed integer nonlinear nonconvex problem. A proximal policy optimization (PPO) based reinforcement learning method is then proposed to determine the AP assignment. The network performance is evaluated by comparing it with exhaustive search, signal strength strategy, and trust region policy optimization methods. The results indicate that the designed method achieves a network-wide data rate with up to 100% and 97.7% optimality in static and mobile user scenarios, respectively.

Index Terms—User-AP Association, Hybrid LiFi/WiFi networking, PPO.

I. INTRODUCTION

With the rapid advancement of the wireless Internet of Things (IoT) and 5G and beyond communication technologies, along with the unprecedented diversification of applications, the number of wirelessly connected devices and the global mobile data traffic is increasing exponentially. A report published by Ericsson predicts that the monthly average usage per smartphone worldwide will reach 19 GB in 2023 [1], with more than 80% mobile data coming from indoor networks such as bandwidth-hungry video streaming. Densely deployed Wireless-Fidelity (WiFi) has been applied to indoor wireless networks to meet the growing demand for broadband connectivity, which poses significant challenges to the existing limited RF spectrum, where massively increasing mobile traffic data cannot be accommodated.

To address the looming radio frequency (RF) spectrum shortage problem, Light-Fidelity (LiFi) [2] based on visible light communication (VLC) is introduced as an alternative wireless technology because: (i) it relies on a large portion of the unregulated spectrum, ranging from 375 THz to 750 THz, which can provide hundreds of terahertz of license-free bandwidth, being approximately 1,300 times that of the radio frequency spectrum (about 300 GHz); (ii) it can be used in RF-restricted environments, such as aircraft and hospital; (iii) it provides inherent secure wireless communication due to the

low-penetration property [3]. LiFi also has some disadvantages compared with RF, including: (i) short coverage range, usually a few meters between the access point (AP) and users; (ii) intermittent link availability because of limited field of view (FoV) and obstructions [4].

Therefore, hybrid LiFi/WiFi networks is proposed to leverage the advantages of fast data transfer of LiFi and wider coverage range of WiFi, thus complementing well with each other to provide a seamless, high data-rate transmission in indoor networks [5]. A number of studies have been conducted with the objective to improve the spectrum efficiency of the hybrid networks, including resource allocation [6], [7], network deployment [8], [9], user-AP association [10], [11], among others. For user-AP association problems in hybrid LiFi/WiFi networks, traditional AP selection methods for homogeneous networks, such as signal strength strategy (SSS) [12] is not applicable as it will make the WiFi APs be prone to overloading¹.

In recent years, a few studies [13]–[19] have been conducted to propose user-AP association methods specifically for the hybrid LiFi/WiFi indoor networks. In [13], the authors propose a two-stage fuzzy logic based AP selection method to determine the user-AP association, where simulation results show that close-to-optimal throughput can be achieved. The authors in [14] assign users to LiFi or WiFi APs based on a predefined data rate threshold, i.e., users with received data rate below the threshold will be connected to the WiFi AP, otherwise connected to the LiFi AP, thus improving data outage rate of standalone LiFi networks. However, these works only consider the indoor network scenarios with static users. A few works [16]–[19] are proposed recently to investigate the user-mobility-aware AP selection methods. For example, a Q-learning method with knowledge transfer is proposed in [16] to select appropriate networks for indoor users by considering the dynamic traffic load overtime at given specific locations, which cannot be directly applied to indoor environment without past learning experience. The work in [17] proposes a reinforcement learning (RL) based AP assignment strategy to maximize the throughput of the hybrid LiFi/WiFi networks while guaranteeing the desired user satisfaction, where random waypoint model (RWP) and hotspot RWP model are used for modeling indoor user moving trajectories, but without

¹In hybrid networks, WiFi has larger coverage area but lower capacity compared to LiFi, thus WiFi AP will be likely to overload when SSS based approaches are adopted.

consideration of practical capacity limitations of the network.

To bridge the gap, in this paper, our objective is to propose a robust user-AP association method for indoor hybrid LiFi/WiFi networks with either static or mobile users, aiming to maximize the achievable sum data rate. Because of the binary AP-user association variables, the resulting formulated problem is a mixed integer nonlinear nonconvex programming problem (MINLP), which is generally NP-hard, where the globally optimal solution cannot be obtained within polynomial time. The paper makes the following main contributions:

- *User-AP association formulation.* We formulate mathematically the user-AP association problem with the objective of maximizing the sum throughput of the hybrid LiFi/WiFi indoor network by jointly considering the network capacity limitation and the user mobility patterns.
- *Proximal policy optimization (PPO) based solution algorithm.* To solve the resulting MINLP problem, we design a solution algorithm based on proximal policy optimization, an actor-critic RL framework that can converge faster than the value-based methods and is more stable than the policy-based methods.
- *Performance evaluation.* The performance of the proposed methods is evaluated in terms of the sum data rate of the hybrid networks by comparing it to the global optimum obtained by exhaustive search (ES), and the results of signal strength strategy and trust region policy optimization (TRPO) methods through extensive simulation experiments.

The remainder of the paper is organized as follows. In Section II, we introduce the system model for the hybrid LiFi/WiFi networks, user-AP association model, and random waypoint model. The centralized optimization problem formulation is present in Section III. Subsequently, we introduce the proposed PPO method in Section IV. Performance evaluation and conclusions are given in Section V and Section VI, respectively.

II. SYSTEM MODEL

We consider an indoor hybrid LiFi/WiFi network as illustrated in Fig. 1(a). The indoor scenario consists of a set of $\mathcal{L} = \{1, 2, \dots, l, \dots, L\}$ LiFi APs and a set of $\mathcal{W} = \{1, 2, \dots, w, \dots, W\}$ WiFi APs. We assume that there is a set of $\mathcal{K} = \{1, 2, \dots, k, \dots, K\}$ active users uniformly distributed over the room, where users can only be served by either one LiFi AP or one WiFi AP. Time division multiple access (TDMA) is adopted for each AP to serve multiple users. From a practical perspective, driven by the finite capacity limitation of the networks, we also limit the maximum number of connected users to each LiFi AP and WiFi AP as N_l and N_w as in [10].

A. LiFi Channel Model

In this study, we consider the line of sight (LoS) downlink path in LiFi networks. Our following analysis can be easily extended to scenarios with LoS and non-LoS paths. The geometry of the LiFi LoS propagation is shown in Fig. 1(b), where the light-emitting diode (LED) transmitter – LiFi AP l is installed on the ceiling, directly facing downwards, and

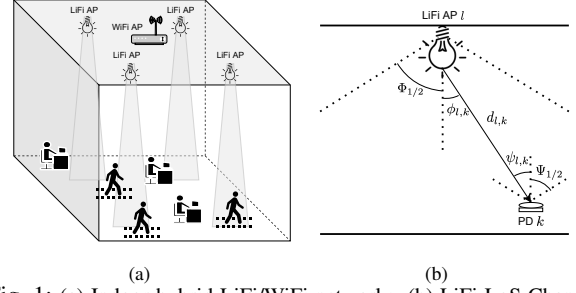


Fig. 1: (a) Indoor hybrid LiFi/WiFi networks, (b) LiFi LoS Channel Model.

the receiving device photodiode (PD) is installed on each user k , facing directly upwards. $d_{l,k}(t)$, $\psi_{l,k}(t)$, and $\phi_{l,k}(t)$ are the distance between the LiFi AP l and the user k , the angle of incidence, and the angle of irradiation at time slot t , respectively. $\Psi_{1/2}$ is the half angle of the PD's FoV, and $\Phi_{1/2}$ is the half-intensity radiation angle of LiFi AP. At time slot t , if user k is served by LiFi AP l , the optical channel gain of the LoS channel can then be given as [20]:

$$H_{l,k}^{lf}(t) = \begin{cases} \frac{(m+1)A}{2\pi d_{l,k}^2(t)} \cos^m(\phi_{l,k}(t)) T_s(\psi_{l,k}(t)) g(\psi_{l,k}(t)) \cos(\psi_{l,k}(t)) & 0 \leq \psi_{l,k}(t) \leq \Psi_{1/2}, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

where m is the Lambertian index and is given by the semi-angle $\Psi_{1/2}$ at half illuminance power of the LED as $m = -\ln 2 / \ln(\cos \Psi_{1/2})$. A is the physical area of the PD. $T_s(\psi_{l,k}(t))$ denotes the gain of an optical filter. $g(\psi_{l,k}(t))$ is the optical concentrator gain at time slot t , given as:

$$g(\psi_{l,k}(t)) = \begin{cases} \frac{n^2}{\sin^2 \Psi_{1/2}} & 0 \leq \psi_{l,k}(t) \leq \Psi_{1/2}, \\ 0 & \text{otherwise,} \end{cases} \quad (2)$$

where n is the refractive index. When user k is connected to LiFi AP l at time slot t , the signal-to-interference-plus-noise ratio (SINR) is calculated as:

$$\gamma_{l,k}^{lf}(t) = \frac{(\kappa H_{l,k}^{lf}(t) P_{lf})^2}{\mathcal{N}_{lf} B_{lf} + \sum_{j \neq l} (\kappa H_{j,k}^{lf}(t) P_{lf})^2}, \quad (3)$$

where κ denotes the optical to electric conversion efficiency of the adopted PD; $H_{l,k}^{lf}(t)$ is the channel gain between the user k and the connected LiFi AP l at time slot t ; P_{lf} and B_{lf} are the transmitted optical power and modulation bandwidth of LiFi APs, respectively; $\mathcal{N}_{lf}$ represents the power spectral density of LiFi noise; $H_{j,k}^{lf}(t)$ is the channel gain between the user k and the interfering LiFi AP j at time slot t . Since the signals in LiFi are non-negative, Shannon capacity cannot be used directly. Instead, we use the lower bound in [21], the achievable data rate between LiFi AP l and user k at time slot t can then be given as:

$$r_{l,k}^{lf}(t) = \frac{B_{lf}}{2} \log_2 \left(1 + \frac{e}{2\pi} \gamma_{l,k}^{lf}(t) \right). \quad (4)$$

B. WiFi Channel Model

For WiFi networks, the path loss model defined in [22] is adopted, where free space loss and shadow fading loss due to large-scale obstructions are considered, given as:

$$L(d_{w,k}(t)) = \begin{cases} L_{FS}(d_{w,k}(t)) + X_{SF} & d_{w,k}(t) \leq d_{BP}, \\ L_{FS}(d_{w,k}(t)) + X_{SF} + 35\log_{10}\left(\frac{d_{w,k}(t)}{d_{BP}}\right) & d_{w,k}(t) > d_{BP}, \end{cases} \quad (5)$$

where $d_{w,k}(t)$ is the distance between the WiFi AP w and the user k at time slot t ; d_{BP} is the breakpoint distance; X_{SF} denotes the shadow fading loss, which is a Gaussian random variable with zero mean and 3 dB deviation if the distance is smaller than or equal to the breakpoint, and 5 dB deviation otherwise; the free space loss L_{FS} is calculated as:

$$L_{FS}(d_{w,k}(t)) = 20\log_{10}(d_{w,k}(t)) + 20\log_{10}(f_c) - 147.5, \quad (6)$$

with f_c denoting the central carrier frequency. At time slot t , the channel transfer function between user k and WiFi AP w can be expressed as:

$$H_{w,k}^{wf}(t) = \sqrt{10^{-\frac{L(d_{w,k}(t))}{10}}}\left(\sqrt{\frac{K}{1+K}}e^{j\phi} + \sqrt{\frac{1}{1+K}}X\right), \quad (7)$$

with K representing the Rician K -factor [22]. For the LoS path, $K = 1$ if $d_{w,k}(t) \leq d_{BP}$, otherwise $K = 0$. ϕ is the angle of arrival or departure of the LoS component. X is a complex Gaussian random variable with zero mean and unit variance. Then, the SINR $\gamma_{w,k}^{wf}(t)$ for the WiFi channel model at time slot t is given as:

$$\gamma_{w,k}^{wf}(t) = \frac{|H_{w,k}^{wf}(t)|^2 P_{wf}}{\mathcal{N}_{wf} B_{wf}}, \quad (8)$$

where $\mathcal{N}_{wf}$ represents the power spectral density of noise of the WiFi networks; P_{wf} and B_{wf} are the transmitted power and modulation bandwidth of each WiFi AP, respectively. The achievable data rate $r_{w,k}^{wf}(t)$ between the WiFi AP w and user k can be calculated using:

$$r_{w,k}^{wf}(t) = \frac{B_{wf}}{2} \log_2(1 + \gamma_{w,k}^{wf}(t)). \quad (9)$$

C. User-AP Association Model

We consider single-home accommodation for users in favor of tractable complexity in modeling and theoretical analysis, i.e., each user is associated with at most one AP, either WiFi AP or LiFi AP. The user association vector $\mathbf{u}_k(t)$ for user k at time slot t is denoted as $\mathbf{u}_k(t) = \{u_{l,k}(t), u_{w,k}(t) | \forall l \in \mathcal{L}, \forall w \in \mathcal{W}\}$, where $u_{l,k}(t)$ and $u_{w,k}(t)$ are binary variables, with $u_{l,k}(t) = 1$ or $u_{w,k}(t) = 1$ when user k is associated with LiFi AP l or WiFi AP w , respectively, 0 otherwise. Then, we can define the user-AP association vector for the entire hybrid network as $\mathbf{u}(t) = \{\mathbf{u}_k(t) | \forall k \in \mathcal{K}\}$.

Once the connections between the users and APs are determined, the sum data rate of the hybrid networks at time slot t can be written as:

$$r_{sum}(t) = \sum_{l=1}^{\mathcal{L}} \sum_{k=1}^{\mathcal{K}} u_{l,k}(t) r_{l,k}^{lf}(t) + \sum_{w=1}^{\mathcal{W}} \sum_{k=1}^{\mathcal{K}} u_{w,k}(t) r_{w,k}^{wf}(t). \quad (10)$$

D. Random Waypoint Model

In this work, we use the random waypoint model [23] to model users' indoor moving trajectories because it can effectively simulate the user's motion patterns in various indoor scenarios, which is widely used by researchers [17], [18]. In the RWP model, each user randomly chooses a destination

point within the deployment region, and then moves toward it along a straight line. Node velocity is also chosen randomly with uniform distribution in the interval of selected minimum and maximum speed. When the node arrives at its destination, it either remains stationary for a predefined pause time or immediately moves again according to the same pattern as above-mentioned. With this model, the mobile users and the stationary users may exist simultaneously at some point.

III. PROBLEM FORMULATION

The network control objective can be stated as maximizing the sum data rate of the hybrid indoor LiFi/WiFi downlink access networks by jointly considering the users' positions, user-AP association variables, as well as user moving patterns subject to bandwidth limitations of the hybrid networks. Define $\mathcal{P}(t) = [\mathcal{P}_1(t), \mathcal{P}_2(t), \dots, \mathcal{P}_K(t)]$ as the user location at time slot t , where height is assumed constant for all users and only x coordinates and y coordinates are considered to be moving over time, with $\mathcal{P}_k(t) = \{x_k(t), y_k(t)\}$. The network control problem can then be formulated as:

Problem: Given: $\mathcal{P}(t)$ (11)

Maximize $f = r_{sum}(t)$ (12)

Subject to: $0 \leq x_k(t) \leq X, \forall k \in \mathcal{K}$, (13)

$0 \leq y_k(t) \leq Y, \forall k \in \mathcal{K}$, (14)

$\mathbf{u}_{l,k}(t) \in \{0, 1\}, \forall k \in \mathcal{K}, \forall l \in \mathcal{L}$, (15)

$\mathbf{u}_{w,k}(t) \in \{0, 1\}, \forall k \in \mathcal{K}, \forall w \in \mathcal{W}$, (16)

$\mathbf{1}^T \mathbf{u}_k(t) \leq 1, \forall k \in \mathcal{K}$, (17)

$\sum_{k \in \mathcal{K}} u_{l,k}(t) \leq N_l, \forall l \in \mathcal{L}$, (18)

$\sum_{k \in \mathcal{K}} u_{w,k}(t) \leq N_w, \forall w \in \mathcal{W}$. (19)

In the above formulated problem, constraints (13) and (14) restrict that all the users should be within the predefined indoor area. Constraints (16) and (15) require that the user-AP association variables are binary. (17) indicates that each user can be only served by one AP. The capacity limitations (aka the maximum number of connected users) of LiFi and WiFi APs are set in (18) and (19).

IV. PROPOSED PPO METHOD

In (12), the individual throughput $r_{l,k}^{lf}(t)$ and $r_{w,k}^{wf}(t)$ are nonconvex/nonconcave functions with respect to location variables. Moreover, the user-AP association variables take only binary values. Therefore the resulting network control problem is a mixed integer nonlinear non-convex programming problem, for which there are in general no existing solution algorithms that can be used to obtain the global optimum in polynomial computational time. In this paper, we propose a solution algorithm based on proximal policy optimization actor-critic RL framework [24], [25], which can achieve globally optimal or nearly-optimal solutions as discussed in Sec. V.

The adopted actor-critic RL framework combines the advantages of the value-based (e.g., Q-learning) and policy-based (e.g., gradient descent) RL methods to achieve faster convergence speed with more stable performance [26], [27]. To overcome the slow parameter update issue in the actor-critic models, the PPO is adopted, where it uses importance sampling technology [28], thus making the sampled data reusable and then further improving the training speed.

As shown in Fig. 2, the agent is composed of two neural networks: actor network and critic network, also called policy and value network [24], respectively. At state s_t , the actor network is used to select actions a_t based on the current policy π_θ (with θ denoting the policy parameter) to interact with the environment, and then receive reward r_{t+1} . Then the experience tuples $\{s_t, a_t, r_{t+1}, s_{t+1}\}$ will be stored in the memory. After several timesteps, the data in the memory will be used to calculate (i) probability ratio between the new and old policies $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$; (ii) generalized advantage estimator $\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \dots + (\gamma\lambda)^{T-t+1}\delta_{T-1}$, with $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, γ being the discount factor, λ representing exponential weight discount, $V(s_t)$ denoting the expected return of the state s_t ; and (iii) squared-error loss $L_t^{VF}(\theta) = \mathbb{E}_t[(V_\theta(s_t) - V_t^{target})^2]$, where $\mathbb{E}_t$ refers to the empirical expectation, and V_t^{target} denotes the target value. Accordingly, the clipped surrogate objective function [25] of the proposed PPO can be given as:

$$L_t^{CLIP}(\theta) = \mathbb{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)], \quad (20)$$

with ϵ being a clipping hyperparameter. The actor and critic network will then be updated by the gradient of $L_t^{CLIP}(\theta)$ and $L_t^{VF}(\theta)$, respectively, thus obtaining the optimal policy.

Next, we present the framework of the proposed RL method in detail as follows.

Agent: The agent is a central unit that can obtain the user positions, and execute the proposed network control algorithm to accomplish the user-AP association.

Actions: The action taken at time slot t is determined by the current policy network, expressed as:

$$a_t = [\mathbf{u}_l(t), \mathbf{u}_w(t)], a_t \in \mathcal{A}, l \in \mathcal{L}, w \in \mathcal{W}. \quad (21)$$

which is composed of the user-AP association vectors of the hybrid networks. $\mathcal{A}$ is the action space including all possible actions.

States: The state at time slot t considers the SINR of each user, given as:

$$s_t = [\gamma(t)], s_t \in \mathcal{S}, \quad (22)$$

where $\gamma(t) = \{\gamma_{l,k}^{lf}(t), \gamma_{w,k}^{wf}(t)\}$, $k \in \mathcal{K}$, $l \in \mathcal{L}$, $w \in \mathcal{W}$ and $\mathcal{S}$ denotes the state space.

Policy: The stochastic policy $\pi_\theta(a_t|s_t)$ is the probability of choosing action a_t at state s_t , which can be defined as:

$$\pi_\theta(a_t|s_t) = \Pr(a_t|s_t). \quad (23)$$

By selecting the actions according to the current policy and interacting with the environment overtime, we then can obtain the state-action sequence trajectory as $\tau = \{s_0, a_0, s_1, a_1, \dots, s_T\}$.

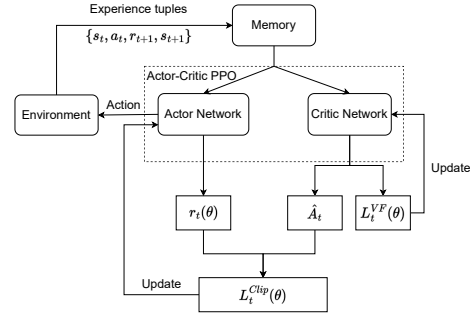


Fig. 2: The framework of the Actor-Critic PPO.

Reward: At time slot t , when action a_t is chosen at state s_t , the immediate reward received is the sum data rate of the networks $r_{sum}(t)$. Therefore, the discounted accumulative reward of a trajectory can be expressed as:

$$R_t = \sum_{t=0}^T \beta^t r_{sum}(t), \quad (24)$$

where $\beta \in (0, 1)$ is the discount factor, which determines the effect of the current immediate reward to the future sum rewards. To ensure that the trained solutions satisfy AP capacity constraints (18) and (19), we assign a larger penalty when the number of users connected to AP exceeds the limitation, thus forcing the agent to avoid infeasible solutions.

TABLE I: LiFi Channel Parameters

Parameter	Symbol	Value
Height between ceiling and user	h	1.5 m - 2 m
The physical area of a PD	A	1 cm ²
The gain of the optical filter	$T_s(\psi_{l,k})$	1.0
Receiver FoV semi-angle	$\Psi_{1/2}$	90 degree
Refractive Index	n	1.5
Half-intensity radiation angle	$\Phi_{1/2}$	60 degree
Detector responsivity	κ	0.53 A/W
Transmit optical power per LiFi AP	P_{lf}	3 Watt
Noise power spectral density	$\mathcal{N}_{lf}$	10 ⁻²¹ A ² /Hz
Bandwidth per LiFi AP	B_{lf}	40 MHz

TABLE II: WiFi Channel Parameters

Parameter	Symbol	Value
Breakpoint distance	d_{BP}	5 m
Central carrier frequency	f_c	2.4 GHz
The angle of arrival/departure of LoS	ϕ	45 degree
Transmit power	P_{wf}	0.1 Watt
Bandwidth per WiFi channel	B_{wf}	10 MHz
Noise power spectral density	$\mathcal{N}_{wf}$	4.002x10 ⁻¹⁷ W/Hz

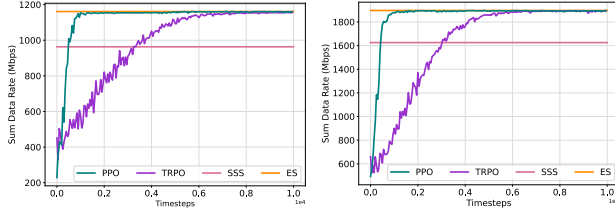
V. PERFORMANCE EVALUATION

In this section, we evaluate the effectiveness of the proposed PPO based solution algorithm through extensive simulations.

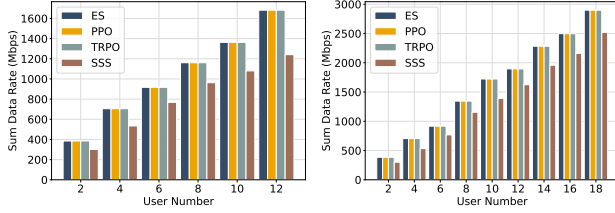
A. Simulation Setup

We consider an indoor room area of $10 \times 10 \times 3.5$ m³, as shown in Fig. 1(a), where $L = 4$ LiFi APs and $W = 1$ WiFi AP² serve $K = \{2, 4, 6, \dots, 18\}$ users. Some other parameters of LiFi and WiFi networks are summarized in Table I and Table II, respectively. To validate the proposed method, we consider two scenarios, i.e., static user and mobile user scenarios. In each scenario, 1 WiFi AP installed at the center of the ceiling and 4 LiFi APs symmetrically deployed

²From the practical perspective, we set the number of WiFi AP to 1, which can be easily extended to arbitrary number as formulated in Sec. II



(a) Network setting 1 with 8 users (b) Network setting 2 with 12 users
Fig. 3: Convergence comparison in static scenarios.



(a) Network setting 1 (b) Network setting 2
Fig. 4: Sum data rate versus user number in static scenarios.

on the whole ceiling are adopted to serve different numbers of users, where two different network settings with different network capacity constraints are considered: (i) Networking Setting 1: $LN_l + WN_w = 13$; (ii) Networking Setting 2: $LN_l + WN_w = 19$. The user location and height are generated randomly. For the RWP model, the user moving speed is randomly selected between 0.5 m/s and 2 m/s. The user location info is obtained every 0.1 s, ensuring that the users' locations are tracked in a nearly real-time manner yet does not change significantly. The learning rate and discount factor for the proposed PPO method are set to 0.0001 and 0.99.

B. Compared Methods

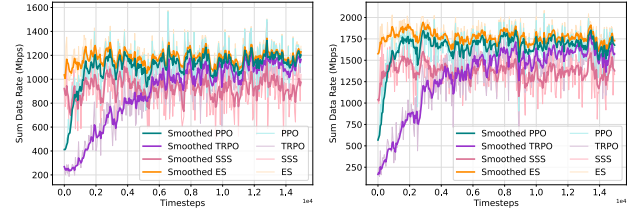
Exhaustive search: Exhaustive search is a brute-force approach to solve optimization problems by evaluating all possible solutions. For the proposed problem, ES is used to enumerate all possible AP-user association cases to find the globally optimal AP-user associations, which will provide the upper bound of our work but at the cost of significant computational complexity.

Signal strength strategy: SSS is a widely adopted technique in wireless communication where it selects the AP that provides the highest signal-to-noise ratio for the users [13], [18].

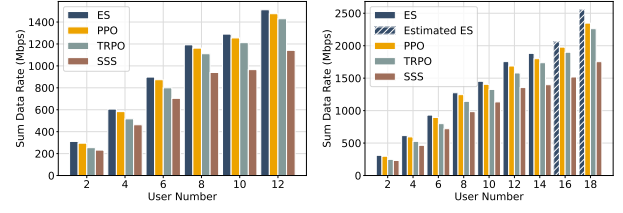
Trust region policy optimization: TRPO is selected to be compared because it has been used to solve user-AP association problem in the hybrid LiFi/WiFi network [17], sharing the same framework with PPO involving actor and critic networks; however, TRPO employs policy updates via trust region constraints, while PPO ensures stability using a clipped function. The learning rate and discount factor for TRPO are the same as those for PPO.

C. Complexity Analysis

As discussed in Sec. IV, the proposed PPO framework consists of actor and critic networks, with each being a 2-hidden-layer deep neural network, where each layer has N_h neurons. The computational complexity of the actor network



(a) Network setting 1 with 8 users (b) Network setting 2 with 12 users
Fig. 5: Convergence comparison in mobile scenarios.



(a) Network setting 1 (b) Network setting 2
Fig. 6: Sum data rate versus user number in mobile scenarios.

is $\mathcal{O}(2(L+W)KN_h + N_h^2)$, while the critic network has a complexity of $\mathcal{O}((L+W)KN_h + N_h^2 + N_h)$. Therefore, the total complexity of the PPO method is $\mathcal{O}(3(L+W)KN_h + 2N_h^2 + N_h)$. The TRPO method has the same complexity as PPO due to the same network architecture. The time complexity of ES and SSS methods for user-AP association are $\mathcal{O}((L+W)K)$ and $\mathcal{O}((L+W)^K)$, respectively.

D. Simulation Results

In this section, the performance of the proposed PPO method is evaluated by comparing it with the results obtained from ES, SSS, and TRPO as above discussed in both static and mobile user scenarios under different network settings.

Static user scenario: Figures 3(a) and 3(b) show the convergence of the proposed PPO and the compared TRPO for 8 static users in network setting 1 and 12 static users in network setting 2, respectively. It can be seen that both PPO and TRPO can converge to the global optimum of the MINLP problem formulated in (12) obtained by ES. However, the proposed PPO converges in a much faster way.

In Fig. 4, we then compare the performance with respect to the sum data rate of the hybrid network of the proposed solution algorithm with the other three methods in the network setting 1 (as in Fig. 4(a)) and network setting 2 (as in Fig. 4(b)) with consideration of different capacity limitations. We can see that the proposed PPO method and TRPO both achieve the global optimal solution as obtained in ES, and outperforms SSS up to 35.5% in both network settings. It can also be observed that as the user number increases, the performance of the proposed PPO will not degrade, thus validating its reliability and scalability.

Mobile user scenario: We then validate the effectiveness in terms of convergence of the proposed PPO algorithm in the mobile scenario, where RWP is used to model the user moving patterns as discussed in Sec. II. The results are shown in Fig. 5, where the smoothed data (i.e., the exponential moving average

of the raw data) is drawn to show the trend in a clearer way. It is obvious that the proposed PPO can not only achieve a nearly optimal performance but also converge faster compared with TRPO.

Next, we further validate the sum data rate performance of the proposed PPO method with mobile users in both network settings with different numbers of users, as shown in Fig. 6. From Fig. 6(a), we can observe that the gaps between PPO and ES are pretty small in all tested cases, at most only up to 5% when the user number is 2. In all of the tested network instances, i.e., when the number of users ranges from 2 to 12, PPO outperforms SSS and TRPO up to 23% and 14%, respectively. Figure 6(b) reports the performance comparison in the network setting 2. For network setting 2 with 16-, and 18-user, the computational complexity of ES increases drastically, where the computer used for simulation can no longer run the ES method to obtain a large number of result sets to get the average results. Therefore, the slash-hatched blue bars are the estimated sum data rate obtained by ES. It can be seen that as the user number increases, the achievable data rate of PPO slightly decreases, with the difference gap only up to 8%, further validating that PPO can reach almost optimal solutions in mobile scenarios. PPO achieves a much better sum data rate compared to SSS and TRPO, up to 25% and 16%, respectively.

VI. CONCLUSIONS

We study and formulate the user-AP association problem in hybrid indoor LiFi/WiFi networks mathematically, where the network control objective is to maximize the network-wide throughput by associating users to appropriate APs with consideration of user indoor mobility patterns and network capacity limitations. The resulting network control problem is an MINLP problem. We then propose a user-AP association algorithm based on proximal policy optimization. Finally, we evaluate the proposed PPO method with extensive simulation results, comparing it with ES, SSS, and TRPO methods. Results show that the proposed solution algorithm can achieve up to around 100% and 97.7% of the global optimum in static and mobile user scenarios, respectively. In future work, we will investigate how to simultaneously and efficiently allocate APs and resources to users to achieve a better quality of service.

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